

Supporting Information

Characterization of Plant Accumulation of Pharmaceuticals from Soils with their Concentration in Soil Pore Water

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Table S1. Therapeutic use and physicochemical properties of the pharmaceuticals used in this study.

Pharmaceutical	Therapeutic use	MW (g/mol)	log K_{ow}	p K_a
Acetaminophen	antipyretic	151.16	0.46	9.38 (acid)
Caffeine	stimulant	194.19	-0.07	—
Carbamazepine	anticonvulsant	236.27	2.45	2.3 (acid), 13.9 (base)
Sulfadiazine	antibacterial	250.28	-0.09	2.01 (base), 6.99 (acid)
Sulfamethoxazole	antibacterial	253.28	0.89	1.6 (base), 5.7 (acid)
Lamotrigine	anticonvulsant	256.10	2.57	5.34 (base)
Carbadox	antibacterial	262.22	-1.22	1.8 (base), 10.5 (acid)
Estrone	estrogen	270.37	3.13	10.77 (acid)
17 β -Estradiol	estrogen	272.38	4.01	10.71 (acid)
Triclosan	antimicrobial	289.54	4.76	7.9 (acid)
Trimethoprim	antibacterial	290.32	0.91	7.12 (base)
Lincomycin	antibacterial	406.54	0.2	7.6 (base)
Oxytetracycline	antibacterial	460.43	-0.9	3.23 (acid), 7.32 (acid), 9.11 (base)
Monensin	antibacterial	692.87	5.43	4.3 (acid)
Tylosin	antibacterial	916.10	1.63	7.73 (base)

Table S2. Mean extraction efficiency (%) of pharmaceuticals from radish tissues and soil with relative standard deviation (%) reported in parenthesis.

Pharmaceuticals	Shoots (%)	Roots (%)	Soil (%)
Acetaminophen	78.0 (4.6)	100.8 (3.2)	73.9 (5.9)
Caffeine	74.6 (5.4)	82.1 (1.6)	71.3 (7.4)
Carbamazepine	88.7 (2.1)	92.4 (5.9)	95.8 (8.1)
Sulfadiazine	73.5 (3.5)	79.1 (3.0)	82.4 (1.4)
Sulfamethoxazole	75.7 (6.1)	80.5 (2.2)	76.7 (3.8)
Lamotrigine	56.4 (4.0)	62.7 (4.6)	68.3 (2.5)
Carbadox	80.6 (3.1)	75.8 (9.3)	97.0 (6.5)
Trimethoprim	76.9 (2.6)	84.3 (6.8)	111.4 (10.1)
Lincomycin	83.9 (7.5)	104.1 (9.7)	81.4 (5.0)
Oxytetracycline	50.6 (5.3)	54.6 (3.8)	39.8 (4.3)
Monensin	86.4 (3.6)	94.8 (10.5)	92.6 (6.2)
Tylosin	60.6 (6.7)	68.7 (2.1)	70.3 (5.8)
Estrone	74.9 (7.7)	80.0 (8.5)	91.8 (11.3)
17 β -Estradiol	73.2 (6.2)	86.1 (9.3)	108.4 (12.4)
Triclosan	85.6 (2.9)	98.4 (7.1)	90.3 (3.7)

Table S3. Method detection limits (MDL) of pharmaceuticals in radish tissues, soils and pore water.

Pharmaceuticals	Leaves (ng/g)	Roots (ng/g)	Soils (ng/g)	Pore water (ng/mL)
Acetaminophen	3.4	3.1	1.5	0.5
Caffeine	1.8	1.6	0.8	0.6
Carbamazepine	0.7	0.7	0.7	0.5
Sulfadiazine	2.7	2.6	2.1	0.5
Sulfamethoxazole	3.4	3.5	3.0	0.5
Lamotrigine	4.2	4.1	3.5	0.5
Carbadox	6.5	6.3	5.2	0.5
Trimethoprim	5.1	4.8	3.7	0.5
Lincomycin	2.4	2.2	1.8	0.7
Oxytetracycline	3.1	3.0	3.3	2.0
Monensin	0.8	0.8	0.7	0.5
Tylosin	2.0	2.1	2.0	0.8
Estrone	2.2	2.0	1.9	1.2
17 β -Estradiol	2.6	2.5	2.2	1.2
Triclosan	4.1	3.8	2.6	1.0

Table S4. Spiked pharmaceutical concentration in soils at the beginning of greenhouse experiment.

Pharmaceuticals	Riddles soil (ng/g)	Capac soil (ng/g)	Spinks soil (ng/g)
Acetaminophen	467 $\pm$ 38	488 $\pm$ 25	471 $\pm$ 18
Caffeine	451 $\pm$ 10	433 $\pm$ 19	454 $\pm$ 7
Carbamazepine	424 $\pm$ 5	430 $\pm$ 8	438 $\pm$ 4
Sulfadiazine	423 $\pm$ 16	488 $\pm$ 35	444 $\pm$ 13
Sulfamethoxazole	464 $\pm$ 30	406 $\pm$ 22	414 $\pm$ 16
Lamotrigine	474 $\pm$ 14	421 $\pm$ 11	455 $\pm$ 30
Carbadox	342 $\pm$ 18	396 $\pm$ 14	363 $\pm$ 9
Trimethoprim	478 $\pm$ 13	493 $\pm$ 19	504 $\pm$ 36
Lincomycin	457 $\pm$ 23	430 $\pm$ 12	413 $\pm$ 29
Oxytetracycline	357 $\pm$ 16	348 $\pm$ 32	318 $\pm$ 38
Monensin	379 $\pm$ 11	413 $\pm$ 8	356 $\pm$ 18
Tylosin	283 $\pm$ 30	339 $\pm$ 15	302 $\pm$ 11
Estrone	492 $\pm$ 26	474 $\pm$ 50	509 $\pm$ 28
17 β -Estradiol	434 $\pm$ 33	392 $\pm$ 12	391 $\pm$ 24
Triclosan	485 $\pm$ 19	515 $\pm$ 26	452 $\pm$ 32

Table S5. The half-life (DT_{50} , days) and the first-order reaction kinetic fitting correlation coefficients (R^2) of pharmaceuticals in three soils.

Pharmaceutical	Riddles soil		Capac soil		Spinks soil	
	DT_{50} (days)	R^2	DT_{50} (days)	R^2	DT_{50} (days)	R^2
Acetaminophen	13.3	0.887	14.1	0.894	14.4	0.889
Caffeine	20.4	0.805	20.4	0.858	30.1	0.955
Carbamazepine	> 100		> 100		> 100	
Sulfadiazine	13.9	0.811	15.8	0.857	18.7	0.922
Sulfamethoxazole	15.1	0.801	15.8	0.879	19.3	0.964
Lamotrigine	46.2	0.946	49.5	0.990	69.3	0.955
Carbadox	46.2	0.942	53.3	0.883	49.5	0.924
Trimethoprim	27.7	0.974	26.7	0.992	43.3	0.908
Lincomycin	9.9	0.852	11.7	0.868	16.9	0.976
Oxytetracycline	28.9	0.841	30.1	0.936	33.0	0.906
Monensin	14.7	0.843	16.1	0.815	21.7	0.823
Tylosin	21.0	0.971	22.4	0.973	43.3	0.959
Estrone	16.1	0.849	18.7	0.790	24.8	0.918
17 β -Estradiol	13.3	0.833	14.7	0.834	18.7	0.818
Triclosan	33.0	0.978	49.5	0.886	115.5	0.933

Table S6. Comparison of soil-pore water distribution coefficient (K_d , mL/g) of pharmaceuticals in the three soils between radish-free soil pots and radish-soils after 30 days of exposure.

Compound	K_d (mL/g)					
	Radish-free soil pots			Radish-soil pots		
	Riddles soil	Capac soil	Spinks soil	Riddles soil	Capac soil	Spinks soil
Acetaminophen	3.0 ± 0.02	4.2 ± 0.22	7.8 ± 0.37	2.8 ± 0.4	4.3 ± 0.5	7.5 ± 0.8
Caffeine	12.5 ± 3.11	18.4 ± 1.84	36.5 ± 0.87	13.1 ± 2.8	18.2 ± 3.1	41.6 ± 6.5
Carbamazepine	2.9 ± 0.10	6.5 ± 0.64	16.4 ± 2.63	3.2 ± 0.5	7.1 ± 1.3	21.6 ± 5.0
Sulfadiazine	1.3 ± 0.26	2.3 ± 0.44	5.3 ± 0.96	1.1 ± 0.2	2.2 ± 0.4	4.6 ± 1.7
Sulfamethoxazole	1.2 ± 0.23	2.0 ± 0.46	4.9 ± 0.43	1.0 ± 0.2	1.7 ± 0.3	4.4 ± 0.7
Lamotrigine	4.7 ± 0.78	16.7 ± 2.27	99.0 ± 20.5	5.3 ± 0.9	15.5 ± 2.9	110.7 ± 22.2
Carbadox	12.4 ± 4.05	35.3 ± 6.49	78.6 ± 14.1	12.7 ± 1.8	34.2 ± 3.6	88.8 ± 22.0
Trimethoprim	31.9 ± 6.32	66.8 ± 1.9	120 ± 10.3	29.8 ± 5.8	50.3 ± 8.5	117.6 ± 19.5
Lincomycin	1.7 ± 0.68	2.6 ± 0.55	5.3 ± 0.73	1.5 ± 0.2	2.7 ± 0.4	5.1 ± 1.0
Oxytetracycline	18.3 ± 7.68	28.0 ± 6.29	30.6 ± 7.83	17.6 ± 3.9	27.8 ± 5.1	33.9 ± 5.3
Monensin	3.4 ± 0.92	5.2 ± 2.08	5.6 ± 1.43	3.3 ± 0.7	4.2 ± 0.6	5.3 ± 0.8
Tylosin	6.1 ± 0.59	9.9 ± 1.51	13.5 ± 3.26	6.6 ± 1.1	11.1 ± 2.0	16.1 ± 2.6
Estrone	36.2 ± 0.94	61.0 ± 16.0	174 ± 77.5	35.6 ± 4.6	62.7 ± 11.5	168.2 ± 14.9
Triclosan	188 ± 23.1	273 ± 10.3	546 ± 130	192.9 ± 30	262.2 ± 25.1	567.7 ± 77.5

Table S7. Linear regression equation and correlation coefficient (R^2) of the relationship of C_{root} vs. C_{pw} and C_{root} vs. C_{soil} .

Pharmaceuticals	C_{root} vs. C_{pw}		C_{root} vs. C_{soil}	
	Regression equation	R^2	Regression equation	R^2
Acetaminophen	$y = 6.06x + 7.75$	0.90	$y = 1.18x + 29.39$	0.11
Caffeine	$y = 5.79x - 4.83$	0.88	$y = 0.32x - 2.04$	0.60
Carbamazepine	$y = 11.67x + 36.29$	0.95	$y = 2.46 + 79.08$	0.31
Sulfadiazine	$y = 0.71x + 5.03$	0.53	$y = 0.14x + 11.44$	0.10
Sulfamethoxazole	$y = 0.51x + 15.45$	0.16	$y = 0.44x + 13.74$	0.28
Lamotrigine	$y = 13.46x + 4.19$	0.82	$y = 1.02x + 292.9$	0.08
Carbadox	$y = 1.36x + 10.89$	0.81	$y = 0.08x + 12.05$	0.42
Trimethoprim	$y = 3.79x + 17.37$	0.82	$y = 0.18x + 5.94$	0.48
Lincomycin	$y = 1.02x + 1.12$	0.97	$y = 0.18x + 2.27$	0.80
Oxytetracycline	$y = 0.80x + 17.69$	0.10	$y = 0.01x + 19.90$	0.03
Monensin	$y = 1.01x - 0.14$	0.93	$y = 0.12x + 1.48$	0.76
Tylosin	$y = 0.26x + 1.35$	0.79	$y = 0.01x + 2.24$	0.41
Estrone	$y = 5.78x + 2.44$	0.19	$y = 0.03x + 3.16$	0.15
Triclosan	$y = 45.09x - 1.98$	0.67	$y = 0.12x + 32.24$	0.32

Table S8. Linear regression equation and correlation coefficient (R^2) of the relationship of C_{shoot} vs. C_{pw} and C_{shoot} vs. C_{soil} .

Pharmaceuticals	C_{shoot} vs. C_{pw}		C_{shoot} vs. C_{soil}	
	Regression equation	R^2	Regression equation	R^2
Acetaminophen	$y = 2.90x + 10.55$	0.65	$y = 0.76x + 15.96$	0.18
Caffeine	$y = 148.5x - 229.31$	0.94	$y = 6.66x - 56.44$	0.48
Carbamazepine	$y = 93.22x + 1624.3$	0.82	$y = 19.0x + 3029.3$	0.24
Sulfadiazine	$y = 1.01x + 4.77$	0.74	$y = 0.06x + 16.59$	0.01
Sulfamethoxazole	$y = 1.29x + 17.18$	0.36	$y = 0.24x + 34.67$	0.04
Lamotrigine	$y = 82.7x - 228.4$	0.82	$y = 6.77x + 1336.1$	0.08
Carbadox	$y = 2.07x + 30.51$	0.58	$y = 0.11x + 35.10$	0.29
Trimethoprim	$y = 18.47x + 100.01$	0.73	$y = 0.86x + 42.09$	0.44
Lincomycin	$y = 1.77x + 13.38$	0.54	$y = 0.32x + 15.22$	0.42
Oxytetracycline	$y = 0.30x + 13.33$	0.03	$y = 0.001x + 14.69$	0.01
Monensin	$y = 0.42x + 0.85$	0.84	$y = 0.05x + 1.46$	0.82
Tylosin	$y = 0.05x + 1.12$	0.70	$y = 0.003x + 1.29$	0.59
Estrone	$y = 0.84x + 3.41$	0.06	$y = 0.03x + 1.09$	0.16
Triclosan	$y = 14.33x + 10.14$	0.61	$y = 0.05x + 17.96$	0.23

Table S9. RCF_{soil} and $RCF_{\text{pore water}}$ values of triclosan accumulation in radish from soils and the soils amended with biosolids or biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{\text{pore water}}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.26 ± 0.06	48.9 ± 14.2	30	This study
Capac soil	2.8	—	0.16 ± 0.02	44.3 ± 7.76	30	This study
Spinks soil	4.9	—	0.09 ± 0.03	50.2 ± 14.7	30	This study
A	0.4	35	2.10 ± 0.10	73.5	35	Fu et al., 2016a, 2016b
B	1.2	137	0.30 ± 0.10	41.1	35	Fu et al., 2016a, 2016b
C	2.5	311	0.25 ± 0.17	77.8	35	Fu et al., 2016a, 2016b
D	3.5	731	0.12 ± 0.04	87.7	35	Fu et al., 2016a, 2016b
E	1.3	—	0.56 ± 0.07	29.8 ± 4.59	35	Li et al., 2020
F	1.4	—	0.54 ± 0.13	31.6 ± 9.63	35	Li et al., 2020
G	1.5	—	0.40 ± 0.06	35.8 ± 1.23	35	Li et al., 2020
H	1.6	—	0.13 ± 0.03	43.8 ± 5.24	35	Li et al., 2019
I	8.0	—	0.43 ± 0.38	—	40	Pannu et al., 2012
J	2.4	—	0.33 ± 0.12	—	34	Prosser et al., 2014
K	1.8	—	1.20 ± 0.40	—	34	Prosser et al., 2014
L	1.0	—	0.12	—	35	Carter et al., 2014

Table S10. RCF_{soil} and $RCF_{\text{pore water}}$ values of acetaminophen uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{\text{pore water}}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	2.94 ± 1.32	8.88 ± 4.04	30	This study
Capac soil	2.8	—	1.65 ± 0.15	6.97 ± 0.75	30	This study
Spinks soil	4.9	—	1.03 ± 0.25	8.05 ± 2.33	30	This study
A	1.3	—	7.34 ± 2.01	8.30 ± 2.23	35	Li et al., 2020
B	1.4	—	4.75 ± 1.53	8.93 ± 1.54	35	Li et al., 2020
C	1.5	—	1.51 ± 0.29	9.32 ± 1.88	35	Li et al., 2020
D	1.6	—	4.48 ± 0.60	6.59 ± 0.08	35	Li et al., 2019

Table S11. RCF_{soil} and $RCF_{pore\ water}$ values of caffeine uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{pore\ water}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.49 ± 0.17	6.27 ± 1.74	30	This study
Capac soil	2.8	—	0.36 ± 0.07	6.10 ± 0.87	30	This study
Spinks soil	4.9	—	0.20 ± 0.05	7.22 ± 1.50	30	This study
A	1.3	—	0.26 ± 0.02	2.66 ± 0.40	35	Li et al., 2020
B	1.4	—	0.29 ± 0.04	3.11 ± 0.49	35	Li et al., 2020
C	1.5	—	0.15 ± 0.01	3.69 ± 1.20	35	Li et al., 2020
D	1.6	—	0.46 ± 0.05	5.18 ± 0.94	35	Li et al., 2019

Table S12. RCF_{soil} and $RCF_{pore\ water}$ values of carbamazepine uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{pore\ water}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	5.05 ± 1.11	14.3 ± 2.70	30	This study
Capac soil	2.8	—	2.04 ± 0.22	13.2 ± 2.27	30	This study
Spinks soil	4.9	—	1.01 ± 0.16	16.2 ± 0.45	30	This study
A	1.3	—	3.07 ± 0.32	11.7 ± 0.88	35	Li et al., 2020
B	1.4	—	2.76 ± 0.91	13.1 ± 2.21	35	Li et al., 2020
C	1.5	—	0.65 ± 0.32	10.4 ± 0.54	35	Li et al., 2020
D	1.6	—	2.71 ± 0.19	11.3 ± 0.83	35	Li et al., 2019
E	2.2	6.9	1.12 ± 0.50	7.75	29	Wu et al., 2012
F	1.0	—	1.05	8.28	35	Carter et al., 2014

Table S13. RCF_{soil} and $RCF_{pore\ water}$ values of sulfadiazine uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{pore\ water}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	1.47 ± 0.50	1.88 ± 0.59	30	This study
Capac soil	2.8	—	0.84 ± 0.10	1.90 ± 0.34	30	This study
Spinks soil	4.9	—	0.37 ± 0.03	1.96 ± 0.35	30	This study
A	1.3	—	2.78 ± 0.45	1.29 ± 0.37	35	Li et al., 2020
B	1.4	—	1.87 ± 0.29	1.37 ± 0.28	35	Li et al., 2020
C	1.5	—	0.28 ± 0.06	1.47 ± 0.80	35	Li et al., 2020
D	1.6	—	1.44 ± 0.16	0.68 ± 0.06	35	Li et al., 2019

Table S14. RCF_{soil} and $RCF_{pore\ water}$ values of sulfamethoxazole uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{pore\ water}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	2.02 ± 0.56	2.57 ± 0.63	30	This study
Capac soil	2.8	—	1.68 ± 0.11	2.99 ± 0.62	30	This study
Spinks soil	4.9	—	0.67 ± 0.13	3.29 ± 0.90	30	This study
A	1.3	—	2.29 ± 0.33	1.30 ± 0.21	35	Li et al., 2020
B	1.4	—	1.38 ± 0.20	1.27 ± 0.15	35	Li et al., 2020
C	1.5	—	0.30 ± 0.11	1.63 ± 0.74	35	Li et al., 2020
D	1.6	—	1.94 ± 0.21	0.70 ± 0.08	35	Li et al., 2019

Table S15. RCF_{soil} and $RCF_{pore\ water}$ values of lamotrigine uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{pore\ water}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	3.67 ± 0.81	16.7 ± 1.52	30	This study
Capac soil	2.8	—	0.96 ± 0.20	16.1 ± 4.48	30	This study
Spinks soil	4.9	—	0.23 ± 0.06	21.7 ± 5.29	30	This study
A	1.3	—	5.98 ± 2.08	13.9 ± 5.77	35	Li et al., 2020
B	1.4	—	4.17 ± 1.13	15.6 ± 5.52	35	Li et al., 2020
C	1.5	—	1.66 ± 0.14	12.1 ± 3.08	35	Li et al., 2020
D	1.6	—	4.79 ± 0.67	11.5 ± 0.37	35	Li et al., 2019

Table S16. RCF_{soil} and $RCF_{pore\ water}$ values of carbadox uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{pore\ water}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.29 ± 0.11	3.34 ± 0.16	30	This study
Capac soil	2.8	—	0.09 ± 0.02	3.06 ± 0.35	30	This study
Spinks soil	4.9	—	0.05 ± 0.02	3.92 ± 1.38	30	This study
A	1.3	—	0.30 ± 0.04	2.11 ± 0.15	35	Li et al., 2020
B	1.4	—	0.28 ± 0.02	2.48 ± 0.27	35	Li et al., 2020
C	1.5	—	0.19 ± 0.04	2.33 ± 0.40	35	Li et al., 2020
D	1.6	—	0.10 ± 0.01	1.46 ± 0.21	35	Li et al., 2019

Table S17. RCF_{soil} and $RCF_{\text{pore water}}$ values of trimethoprim uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{\text{pore water}}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.32 ± 0.10	10.4 ± 3.85	30	This study
Capac soil	2.8	—	0.17 ± 0.04	8.21 ± 0.82	30	This study
Spinks soil	4.9	—	0.06 ± 0.01	7.08 ± 0.95	30	This study
A	1.3	—	0.30 ± 0.02	7.48 ± 3.91	35	Li et al., 2020
B	1.4	—	0.25 ± 0.04	7.97 ± 0.76	35	Li et al., 2020
C	1.5	—	0.18 ± 0.03	8.45 ± 1.50	35	Li et al., 2020
D	1.6	—	0.21 ± 0.02	4.25 ± 0.28	35	Li et al., 2019

Table S18. RCF_{soil} and $RCF_{\text{pore water}}$ values of lincomycin uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{\text{pore water}}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.78 ± 0.10	1.35 ± 0.32	30	This study
Capac soil	2.8	—	0.48 ± 0.17	1.23 ± 0.50	30	This study
Spinks soil	4.9	—	0.20 ± 0.03	1.07 ± 0.05	30	This study
A	1.3	—	0.99 ± 0.25	1.85 ± 0.40	35	Li et al., 2020
B	1.4	—	0.42 ± 0.07	1.56 ± 0.19	35	Li et al., 2020
C	1.5	—	0.14 ± 0.02	1.19 ± 0.11	35	Li et al., 2020
D	1.6	—	0.30 ± 0.05	1.42 ± 0.18	35	Li et al., 2019

Table S19. RCF_{soil} and $RCF_{\text{pore water}}$ values of oxytetracycline uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K_d (g/mL)	RCF_{soil} (g/g)	$RCF_{\text{pore water}}$ (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.45 ± 0.10	7.85 ± 2.88	30	This study
Capac soil	2.8	—	0.24 ± 0.10	6.59 ± 2.36	30	This study
Spinks soil	4.9	—	0.25 ± 0.06	7.59 ± 2.74	30	This study
A	1.3	—	0.23 ± 0.06	5.77 ± 1.81	35	Li et al., 2020
B	1.4	—	0.17 ± 0.04	6.01 ± 0.75	35	Li et al., 2020
C	1.5	—	0.14 ± 0.02	5.89 ± 0.49	35	Li et al., 2020
D	1.6	—	0.04 ± 0.01	1.63 ± 0.19	35	Li et al., 2019

Table S20. RCF_{soil} and RCF_{pore water} values of monensin uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K _d (g/mL)	RCF _{soil} (g/g)	RCF _{pore water} (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.26 ± 0.08	0.89 ± 0.23	30	This study
Capac soil	2.8	—	0.20 ± 0.04	0.98 ± 0.24	30	This study
Spinks soil	4.9	—	0.19 ± 0.07	1.02 ± 0.34	30	This study
A	1.3	—	0.19 ± 0.02	1.03 ± 0.18	35	Li et al., 2020
B	1.4	—	0.16 ± 0.04	1.03 ± 0.23	35	Li et al., 2020
C	1.5	—	0.11 ± 0.01	0.98 ± 0.07	35	Li et al., 2020
D	1.6	—	0.15 ± 0.05	1.00 ± 0.14	35	Li et al., 2019

Table S21. RCF_{soil} and RCF_{pore water} values of tylosin uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K _d (g/mL)	RCF _{soil} (g/g)	RCF _{pore water} (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.07 ± 0.02	0.49 ± 0.19	30	This study
Capac soil	2.8	—	0.05 ± 0.00	0.45 ± 0.10	30	This study
Spinks soil	4.9	—	0.03 ± 0.01	0.43 ± 0.24	30	This study
A	1.3	—	0.03 ± 0.01	0.34 ± 0.05	35	Li et al., 2020
B	1.4	—	0.02 ± 0.01	0.23 ± 0.11	35	Li et al., 2020
C	1.5	—	0.01 ± 0.00	0.32 ± 0.13	35	Li et al., 2020
D	1.6	—	0.02 ± 0.00	0.32 ± 0.05	35	Li et al., 2019

Table S22. RCF_{soil} and RCF_{pore water} values of estrone uptake by radish from soils and the soils amended with biochars.

Soil	Organic carbon (%)	K _d (g/mL)	RCF _{soil} (g/g)	RCF _{pore water} (mL/g)	Exposure (day)	References
Riddles soil	1.3	—	0.19 ± 0.05	6.92 ± 1.54	30	This study
Capac soil	2.8	—	0.14 ± 0.01	8.95 ± 1.94	30	This study
Spinks soil	4.9	—	0.04 ± 0.01	8.03 ± 4.50	30	This study
A	1.3	—	0.09 ± 0.01	9.35 ± 1.42	35	Li et al., 2020
B	1.4	—	0.05 ± 0.02	9.61 ± 2.68	35	Li et al., 2020
C	1.5	—	0.02 ± 0.00	8.41 ± 1.32	35	Li et al., 2020
D	1.6	—	0.08 ± 0.01	22.1 ± 2.60	35	Li et al., 2019

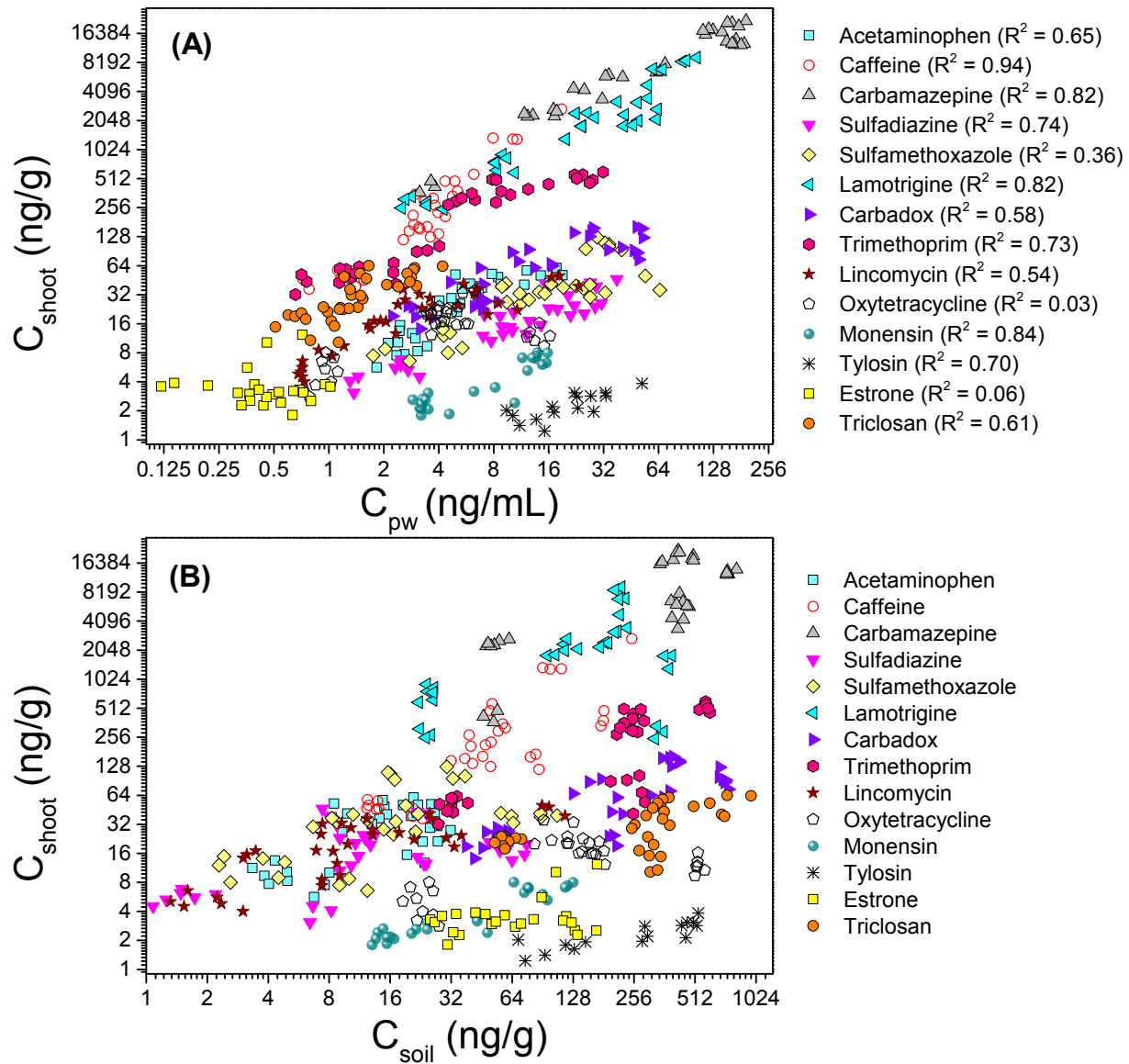


Figure S1. (A) Relation between pharmaceutical concentration in radish shoots (C_{shoot}) and their concentration in soil pore water (C_{pw}). (B) Relation between C_{shoot} and pharmaceutical concentration in bulk soil (C_{soil}). For each pharmaceutical there are the total of 33 data pairs including $n = 9$ from this study plus the remaining from Li et al. (2019 and 2020), except for tylosin ($n = 15$), monensin ($n = 24$), and estrone ($n = 24$). n is the number of data pairs present in the figure.

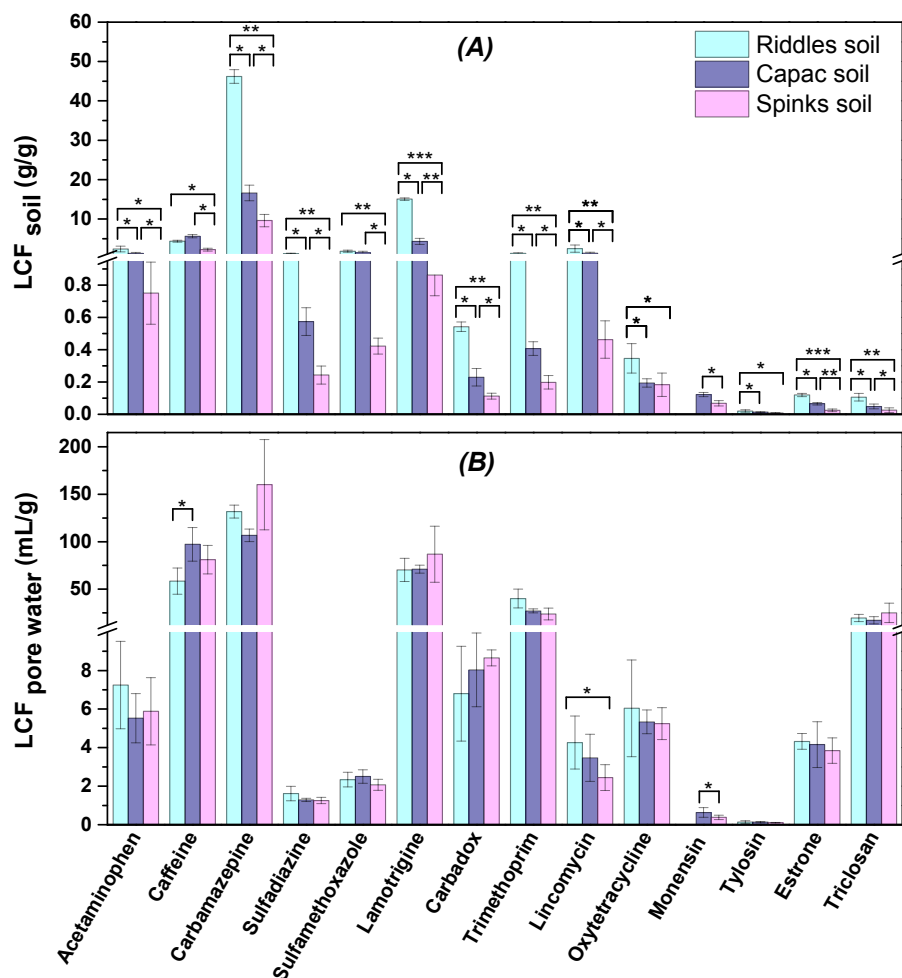


Figure S2. (A) Soil-based leaf concentration factor (LCF_{soil}) and (B) pore water-based leaf concentration factor ($LCF_{pore\ water}$) of pharmaceutical accumulation by radish from soils. Error bars are standard deviations ($n = 3$). *, **, and *** represent the significant difference of LCF values between two soils at $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

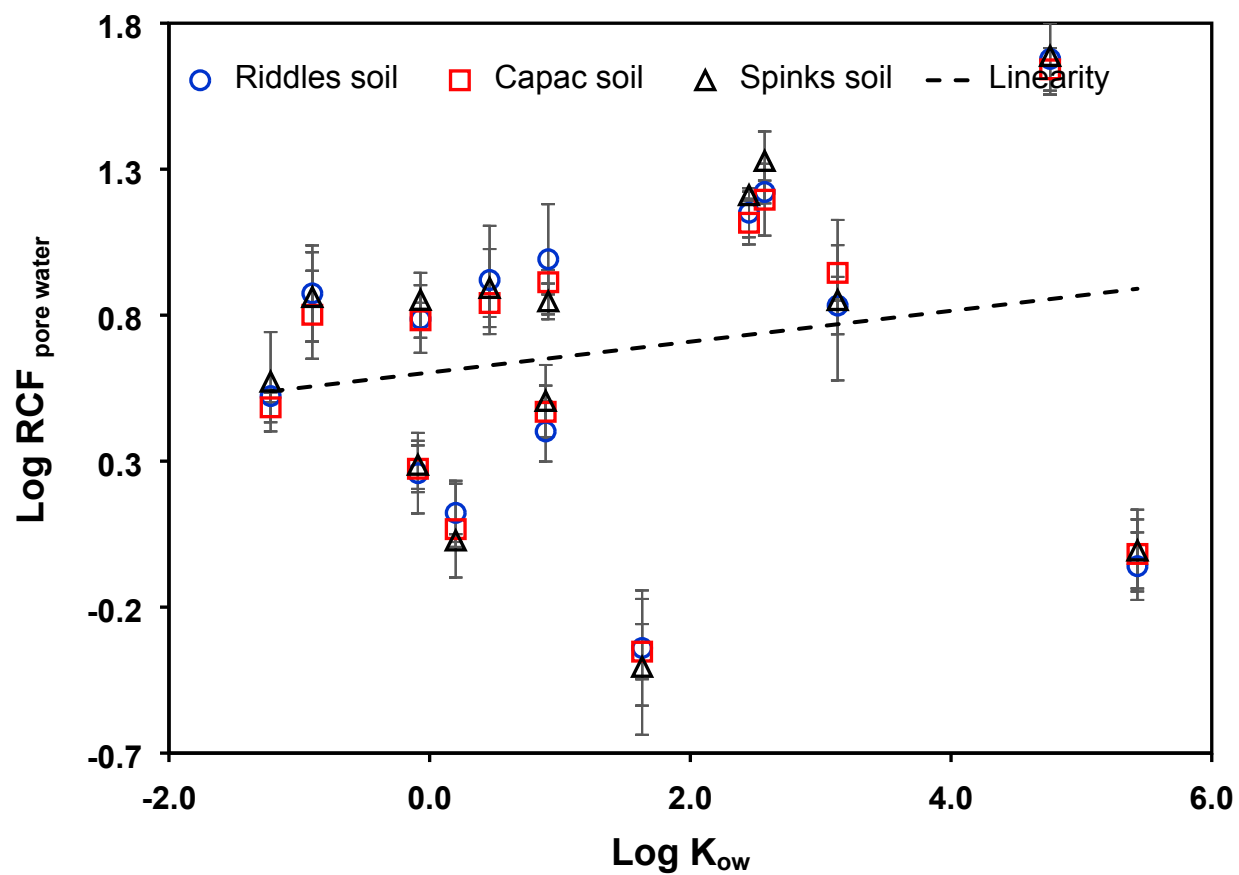


Figure S3. Linear relationship between $\log \text{RCF}_{\text{pore water}}$ and $\log K_{\text{ow}}$ for all measured pharmaceuticals (encompassing neutral, acid and basic pharmaceuticals).

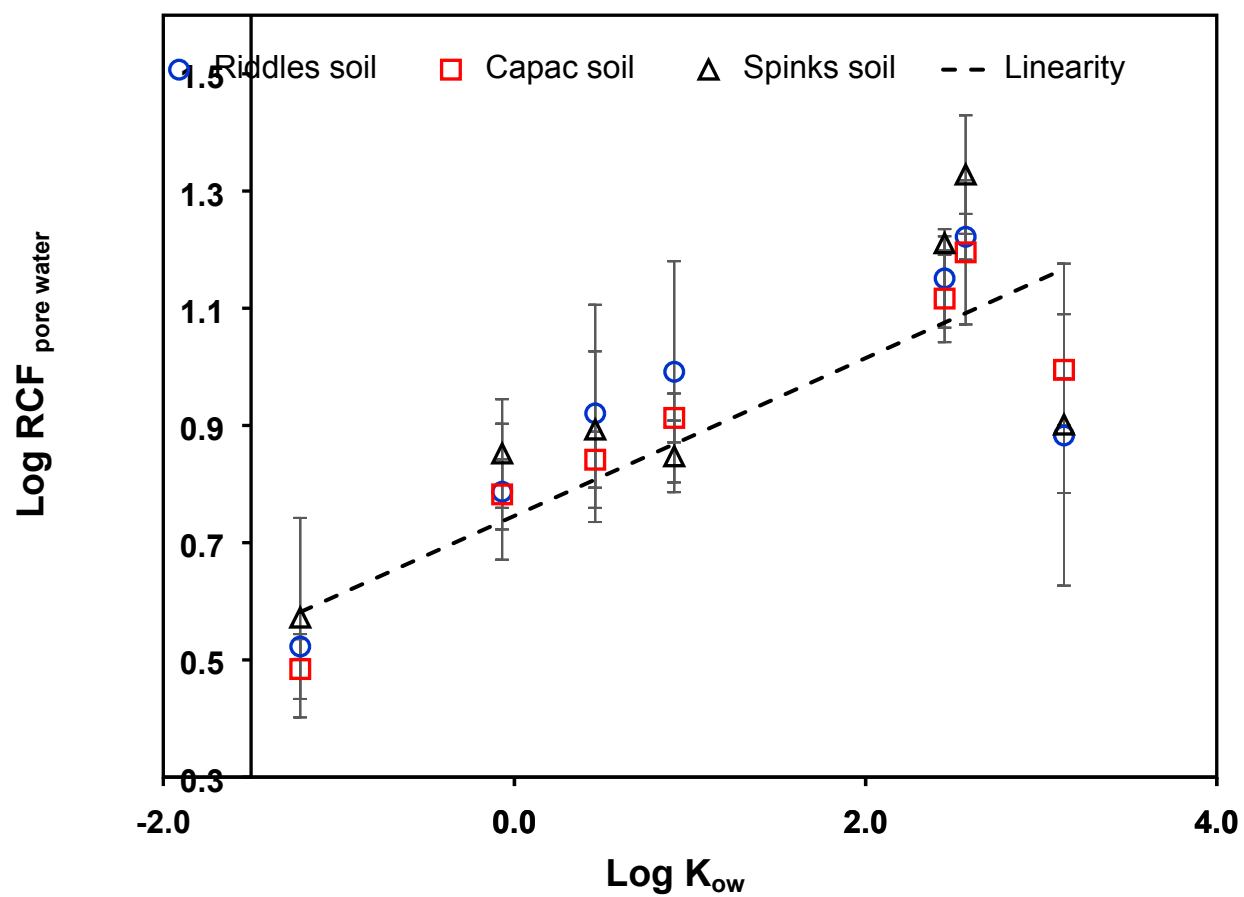


Figure S4. Linear relationship between $\log \text{RCF}_{\text{pore water}}$ and $\log K_{\text{ow}}$ for the neutral pharmaceuticals (acetaminophen, caffeine, carbamazepine, lamotrigine, carbadox, trimethoprim and estrone).